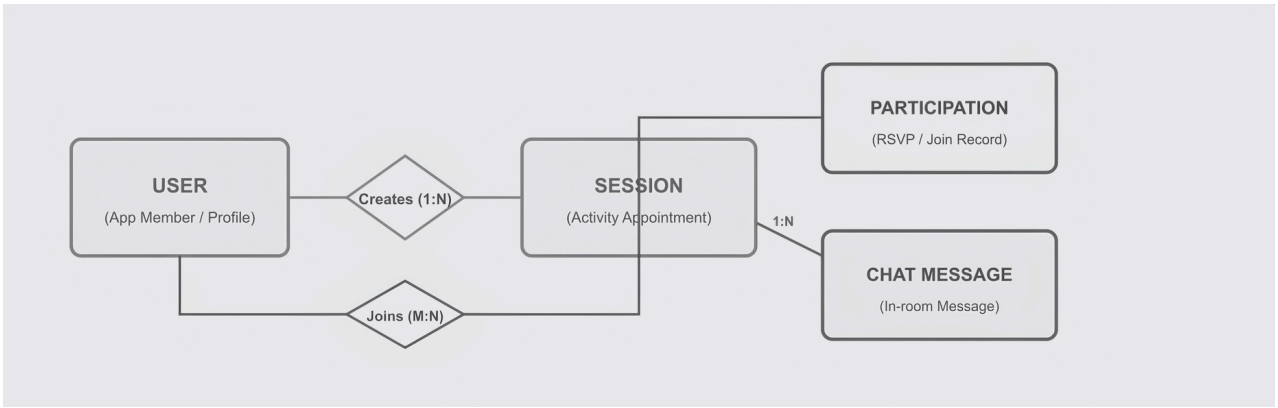
NUDTEE - Data Model Specification

Conceptual & Logical Data Architecture

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1. Conceptual Data Model (Entity Concept Diagram)



2. Logical Data Model & Relationship Rules

Relationship Pair	Cardinality	Foreign Keys	Business Logic Constraints & Rules
USER → SESSION (Organizer)	1 : N (One to Many)	**FK** sessions.organizer_id → users.id	A user can organize multiple sessions, but each session has exactly 1 primary organizer (FR-1).
USER ↔ SESSION (Participant)	M : N (Many to Many)	Resolved via associative entity participations **FK** session_id, **FK** user_id	A user can join multiple sessions, and a session hosts multiple players up to max_players capacity (FR-2).
SESSION → CHAT (In-room Chat)	1 : N (One to Many)	**FK** chat_messages.session_id → sessions.id	A session contains multiple chat messages. Messages are visible exclusively to confirmed room attendees (FR-5).
USER → CHAT (Sender)	1 : N (One to Many)	**FK** chat_messages.sender_id → users.id	A joined participant can send multiple chat messages within an authorized session room (FR-5).

3. Physical Data Model (Physical Attributes & Data Types)

Table Name: users (User Profiles)

Attribute	Data Type	Constraints	Description
id	UUID	**PK** Not Null	Primary Key mapped directly to Supabase Auth (`auth.users.id`)
display_name	VARCHAR(100)	Not Null	User display name synchronized from Google OAuth
email	VARCHAR(255)	Not Null, Unique	User email address used for authentication

Table Name: sessions (Activity Sessions)

Attribute	Data Type	Constraints	Description
id	UUID	**PK** Default gen_uuid()	Primary Key for the session room
organizer_id	UUID	**FK** Not Null	Foreign Key referencing `users.id` (Session Creator)
title / location_name	VARCHAR / TEXT	Not Null	Activity title and venue location text description

latitude / longitude	DECIMAL(10,8)	Not Null	Map coordinates for Leaflet / OpenStreetMap pin rendering (FR-4)
activity_datetime	TIMESTAMPTZ	Not Null	Scheduled activity start date and time
closing_time	TIMESTAMPTZ	Not Null, Check	Registration deadline (must be before `activity_datetime`) (FR-6)
min_players / max_players	INTEGER	Check (max >= min)	Capacity controls for minimum and maximum participants (FR-1, FR-2)
status	VARCHAR(20)	Default 'Open'	Session status: 'Open', 'Full', or 'Closed' (FR-3, FR-6)

Table Name: participations (Participation Bridge Entity)

Attribute	Data Type	Constraints	Description
id	UUID	**PK** Default gen_uuid()	Primary Key for the participation record
session_id / user_id	UUID	**FK** Unique Pair	Foreign Keys mapping user to joined session; prevents duplicate joins (FR-2)
joined_at	TIMESTAMPTZ	Default now()	Timestamp when player tapped Join

Table Name: chat_messages (In-room Messages)

Attribute	Data Type	Constraints	Description
id	UUID	**PK** Default gen_uuid()	Primary Key for chat message
session_id / sender_id	UUID	**FK** Not Null	Foreign Keys linking message to session room and sender (FR-5)
message / created_at	TEXT / TIMESTAMPTZ	Not Null, Default now()	Message content delivered in real time within 2 seconds (FR-5, NFR-1)